

Stephen Zhang

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EDUCATION

Stanford University

Bachelor of Science in Mathematics and Computer Science

Stanford, CA

Expected June 2028

Minor: Music (Trumpet Performance)

Relevant Coursework: Artificial Intelligence, Statistical and Machine Learning, Algorithm Design, Partial and Stochastic Differential Equations, Real Analysis, Probability Theory, Discrete Mathematics

EXPERIENCE

Alpha AI Engineering

AI Engineering Intern

Austin, TX

June 2026 – August 2026

- Fine-tuned a multi-billion-parameter language model with LoRA through staged supervised fine-tuning and direct preference optimization, grading outputs by execution rather than by model scoring
- Ran transformer pretraining sweeps in PyTorch on a SLURM cluster, owning corpus generation, checkpointing, and evaluation
- Extended the Rust scheduling backend of an Anki fork with a SvelteKit interface, self-hosted sync server, and Android build
- Built full-stack TypeScript and React applications on Firebase, using Firestore, Cloud Functions, and federated authentication

EquityBound

Intern

Remote

January 2026 – June 2026

- Valued small and medium enterprises using standard financial methods
- Proposed and implemented service enhancements with the engineering team
- Audited the codebase and identified accuracy and performance improvements

RESEARCH EXPERIENCE

Feldman Lab, Department of Biology, Stanford University

Research Intern

Stanford, CA

June 2023 – Present

- Build and analyze stochastic, agent-based, and equation-based models of gene-culture coevolution in Python (NumPy, SciPy)
- Derive analytical results and validate them against simulations
- Published one [peer-reviewed paper](#) as second author, with a second manuscript under review

PUBLICATIONS

- Laurel Fogarty, **Stephen Zhang**, Marcus W. Feldman (2025). [Gene-culture association and coevolution](#). *Theoretical Population Biology*, Vol. 165, pp. 62–71
- Smarties Sandwiches (2024). *Pi in the Sky*, Pacific Institute for the Mathematical Sciences (PIMS), [issue 23](#), pp. 24–27

SKILLS

- **Languages:** Python, C++, Java, TypeScript, Rust, R, SQL, LaTeX
- **Machine learning:** PyTorch, Hugging Face, LoRA, direct preference optimization, NumPy, SciPy, SLURM
- **Web and cloud:** React, SvelteKit, Firebase (Firestore, Cloud Functions), AWS, Git
- **Spoken languages:** English (native), Mandarin Chinese (fluent reading and writing)

LEADERSHIP AND ACTIVITIES

Stanford Orchestras

Financial Officer

Stanford, CA

September 2025 – Present

- Oversee orchestra operations on the executive committee, alongside the co-presidents and vice president
- Manage a budget of more than \$40K, funding every event while holding spending under plan

AWARDS

- USA Physics Olympiad Semifinalist (Top 400 nationally) 2022
- 5x AIME Qualifier 2020 – 2025
- Stanford Math Tournament – Distinguished Honorable Mention (Top 10%) 2024
- Fellow of the Royal Schools of Music in Trumpet Performance 2025
- CLASS National Chinese Essay Contest Gold Prize 2023